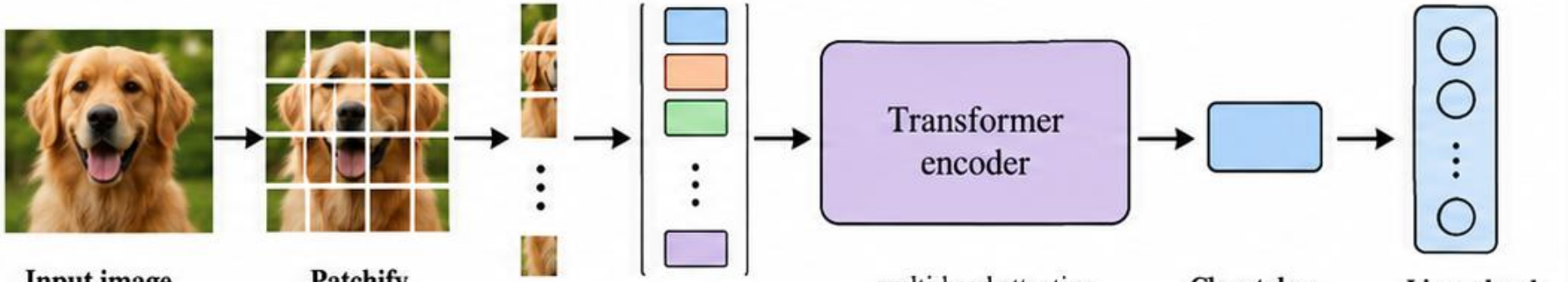
Vision Transformer pipeline for image classification

Conceptual architecture: image patches become tokens processed by global self-attention.



Input image
224 × 224 × 3
(RGB)

Patchify
non-overlapping patches
(e.g., 16 × 16)

Token embedding
patch + position + class
(e.g., 197 × 768)

multi-head attention
+ feed-forward network
(e.g., L × 768)

Class token
final representation
(1 × 768)

Linear head
class probabilities
(e.g., 1000 classes)

How spatial evidence is handled

Self-attention lets token pairs exchange information; position embeddings supply location information. Token count grows as patches become smaller, so attention cost and resolution must be chosen together.